

Human emotions play an important role in individual and group behaviors, influencing both prosocial and antisocial exchanges. Complex and personal, emotions are difficult to study. Although physiological indicators of emotion, such as pulse or skin conductance, are easy to measure, the *subjective experience* of emotion is much harder to assess.

Further complicating the issue, there is no universally accepted theory regarding the role of physiological processes in the experience of emotion. Common sense would suggest that emotion is experienced first, followed by physiological and behavioral events: we feel scared and then break into a cold sweat. However, one of the earliest modern theories of emotion proposed just the opposite—emotion occurs *after* a physiological response, not before: we break into a cold sweat *and then feel scared*.

Males and females also appear to experience emotion differently; however, it is unclear whether this is the result of nature or nurture. Studies using functional magnetic resonance imaging (fMRI) suggest a biological difference. Data suggest that compared to females, males demonstrated greater activation of the amygdala while viewing images meant to induce fear, anger, or sexual arousal, but when asked to suppress negative emotions, females demonstrated greater activation of the prefrontal cortex than males. Research also suggests that females may demonstrate more empathy; when watching another person experience an emotion (eg, pain), females show more neural activation of the same areas that are active during the personal experience of that emotion.

Social factors also appear to play an important role in gender differences. One study, which assessed how boys and girls express emotion from infancy to adolescence, concluded that gender differences appear to emerge around age 2–4 and become more pronounced with age. The researchers hypothesized that children learn gendered behavior regarding emotion through social learning. Cultural differences in the expression of emotion lend support to this argument: several studies suggest that, in general, men are expected to express less emotion than women, with the exception of anger.

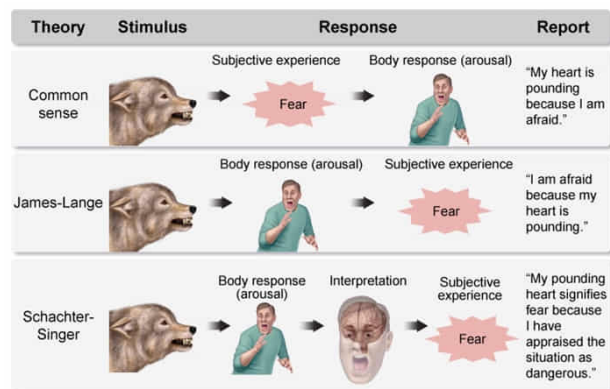
Which of the following theories is most consistent with the theory described in the second paragraph?

- ☐ A. Evolutionary [2%]
- ☐ B. Cognitive appraisal [4%]
- ☐ C. Schachter-Singer [15%]
- ✔ ☒ D. James-Lange [78%]

Correct

79% answered correctly

Explanation:



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The commonsense view of emotion suggests that something (eg, seeing an angry dog) first causes the emotion (eg, fear) to be experienced; then the emotion directly causes a physiological response (eg, increased heart rate).

According to the **James-Lange theory** of emotion, the sequence is reversed. This theory suggests that something (eg, dog) **first** causes a **physiological response** (eg, increased heart rate), which then causes the brain to register the associated emotion (eg, fear). In other words, *specific physiological reactions produce specific emotions*.

The theory described in the second paragraph suggests that emotion occurs *after* a physiological response, which is most consistent with the James-Lange theory of emotion.

(Choice A) According to the evolutionary theory, emotions evolved just like physical traits for the purpose of adapting to the environment. For example, fear evolved as a way to increase arousal when responding to environmental threats. The theory described in the second paragraph does not discuss the *purpose* of emotion.

(Choice B) According to the cognitive appraisal theory, evaluation of the situation (appraisal) precedes emotional and physiological reactions and determines what emotion will be experienced. If appraised as negative (eg, skiing is scary), the emotional response will be negative (eg, fear). If appraised as positive (eg, skiing is fun), the emotional response will be positive (eg, excitement). The second paragraph does not mention cognitive appraisal.

(Choice C) According to the Schachter-Singer theory, emotion is the result of physiological arousal *and* cognitive appraisal. Like the James-Lange theory, physiological arousal is experienced first. Unlike the James-Lange theory,

environmental cues are then used to interpret this arousal to assign an appropriate emotion. The second paragraph does not mention cognitive appraisal.

Educational objective:

The commonsense view of emotion suggests that emotion (eg, fear) causes physiological responses (eg, racing heart, sweating). The James-Lange theory suggests that specific physiological responses (eg, racing heart, sweating) produce specific emotions (eg, fear).

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401022

System:
Motivation, Emotion, Attitudes, Personality, and Stress

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Social factors also appear to play an important role in gender differences. One study, which assessed how boys and girls express emotion from infancy to adolescence, concluded that gender differences appear to emerge around age 2–4 and become more pronounced with age. The researchers hypothesized that children learn gendered behavior regarding emotion through social learning.

Cultural differences in the expression of emotion lend support to this argument: several studies suggest that, in general, men are expected to express less emotion than women, with the exception of anger.

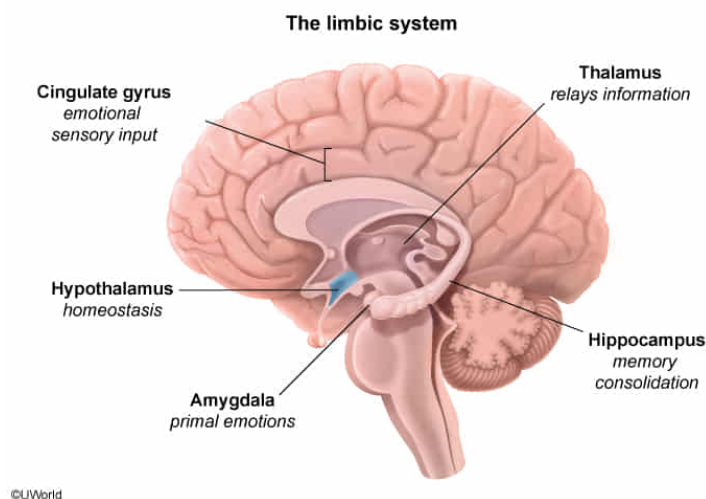
According to the early theory of emotion described in the second paragraph, which of the following brain structures is most critical to the experience of emotion?

- ☐ A. Prefrontal cortex [26%]
- ✓ ☒ B. Hypothalamus [48%]
- ☐ C. Hippocampus [21%]
- ☐ D. Posterior pituitary [3%]

Correct

49% answered correctly

Explanation:



The **limbic system** is a network of brain regions involved in **emotion**, **learning**, and **memory**. The limbic system includes the amygdala, hippocampus, hypothalamus, and cingulate gyrus.

The **hypothalamus** regulates the **pituitary gland** and the **autonomic nervous system**. It controls the release of hormones that maintain homeostatic functions (parasympathetic "rest-and-digest" response) or prepare the body to respond to a stressor (sympathetic "fight-or-flight"

response). The hypothalamus is responsible primarily for the **physiological component of emotion**, such as changes in heart or respiration rate.

The James-Lange theory of emotion (second paragraph) suggests that *specific physiological reactions produce specific emotions*. For example, if experiencing increased heart rate and sweating, one will experience fear. Therefore, the hypothalamus is most critical to the experience of emotion according to the James-Lange theory.

(Choice A) The prefrontal cortex plays a role in what is thought to be a uniquely human capability: the modulation of emotions. In other words, the prefrontal cortex helps regulate emotional responses, such as dampening anger so that, for example, lashing out at bosses does not occur. The modulation of emotion involves cognitive appraisal of the situation, which is not an aspect of the James-Lange theory.

(Choice C) The hippocampus is involved primarily in memory consolidation, which is not a critical component of the James-Lange theory.

(Choice D) The posterior pituitary, a collection of hypothalamic axonal projections, releases the hormones oxytocin and vasopressin. Oxytocin is involved primarily in attachment and pair bonding. Vasopressin, or antidiuretic hormone, regulates the reabsorption of water from the kidneys. Neither hormone appears to be as critical to the experience of emotion as those involved in the fight-or-flight response.

Educational objective:

The limbic system comprises the amygdala, hippocampus, hypothalamus, and cingulate gyrus and is responsible for emotion, learning, and memory. The hypothalamus influences the physiological component of emotion by regulating the pituitary gland and the autonomic nervous system.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401023

System:
Motivation, Emotion, Attitudes, Personality, and
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Which of the following conclusions is most supported by information in the third paragraph?

- ☐ A. The entire limbic system appears to be more active in males than in females. [12%]
- ☐ B. The prefrontal cortex is more likely to be active in males when expressing anger. [5%]
-  ☒ C. Vicarious emotions are associated with greater activation of the mirror neuron system in females. [68%]
- ☐ D. Suppression of negative emotions is associated with activation of the limbic system in females. [13%]

Correct

68% answered correctly

Explanation:

Mirror neurons are a type of neuron that fires when an individual both **observes** and **performs** a behavior; they may play a central role in observational learning and the experience of vicarious emotions. Mirror neurons appear to be involved in grasping the intentions and in understanding the emotions of others. Deficiencies in the mirror neuron system are hypothesized to play a role in autism spectrum disorders.

Vicarious emotions occur when an observer **intuits the feelings** that another might be experiencing. Vicarious emotions are thought to have evolved to help individuals relate to and understand one another through shared emotional expression. fMRI data suggest that neural activation when one experiences an emotion is similar to when one *watches someone else experience that emotion*, suggesting that vicarious emotions are linked to the mirror neuron system.

The third paragraph suggests that vicarious emotions may be associated with greater activation of the mirror neuron system in females than in males.

(Choice A) The third paragraph states that "males demonstrated greater activation of the amygdala while viewing images meant to induce fear, anger, or sexual arousal." While the amygdala is a component of the limbic system, this does not support the conclusion that the *entire* limbic system appears to be more active *overall* in males than in females.

(Choice B) The third paragraph states that "when asked to suppress negative emotions, females demonstrated greater activation of the prefrontal cortex than males." This does not support the conclusion that the prefrontal cortex is more likely to be active in males when expressing anger.

(Choice D) There is no information in the third paragraph to suggest that suppression of negative emotions is associated with activation of the limbic system in females.

Educational objective:

Mirror neurons, which fire both during observation and performance of a behavior, are believed to play a role in the experience of vicarious emotions. Vicarious emotions involve the ability to intuit how someone else feels and are thought to encourage prosocial behaviors through shared emotional expression.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401024

System:
Motivation, Emotion, Attitudes, Personality, and Stress

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This passage considers how gender differences in emotion may result from all of the following, EXCEPT:

- ✔ ☒ A. social learning overriding genetic predisposition starting around age 2–4. [33%]
☐ B. socialization across an individual's lifespan. [40%]
☐ C. social and cultural norms that differ for men and women. [3%]
☐ D. physiological differences between the brains of males and females. [21%]

Correct

34% answered correctly

Explanation:

In Western cultures (eg, North America, Europe), women are more likely than men to express all emotions, except anger, which men are more likely to express. Studies suggest that women are also better than men at detecting and correctly interpreting emotional cues (eg, facial expressions, body language) in others.

Data indicate that **gender differences** in the **expression** and **detection of emotion** are caused by a combination of biological and sociocultural factors, including:

- **Biological differences:** Studies suggest a correlation between anger/aggression and testosterone levels.
- **Social differences:** In Western cultures, girls are socialized to be more emotionally expressive than boys and are encouraged to seek emotional support during stressful situations. Conversely, boys are not encouraged to express emotions that will make them appear "weak" and are encouraged to deal with stress with goal-directed problem-solving.
- **Cultural differences:** In Western cultures, which tend to value individualism, emotional expression is more acceptable than in non-Western cultures, which value collectivism. Differences in the expression of emotion between women and men tend to be more pronounced in Western cultures, which stereotype women as "emotional" and men as "rational."

This passage does not discuss how social learning *overrides* genetic predisposition starting around age 2–4, since the passage suggests that gender differences in emotion are the result of *both* biological and sociocultural factors.

(Choice B) The fourth paragraph describes a study concluding that gender differences appear to emerge around age 2–4 and become more pronounced with age. In other words, gender differences in emotion may result from the lifelong process of socialization, which involves learning society's norms, values, and expected behaviors.

(Choice C) The fourth paragraph describes how children learn gendered behavior through social learning, which is an example of social norms (cultural rules governing behavior) influencing the expression of emotion.

(Choice D) The third paragraph describes differences in brain activity between males and females while experiencing different emotions. Therefore, this passage considers how gender differences in emotion may result from physiological differences between the brains of males and females.

Educational objective:

Gender differences in the expression and detection of emotion are thought to result from both biological and social factors. Biological factors such as genetics may result in men and women processing emotion differently. Social factors such as norms may result in women and men learning to express and interpret emotion differently.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401025

System:
Motivation, Emotion, Attitudes, Personality, and Stress

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Which interpretation of the information in the passage is LEAST consistent with feminist theory?

- ☐ A. Strict gender roles that encourage boys to suppress their emotions are oppressive. [9%]
- ☐ B. Women are often penalized for expressing certain emotions, whereas men are rewarded for expressing them. [5%]
- ☒ C. Differences in emotional expression between men and women help maintain equilibrium in society. [81%]
- ☐ D. Traditional gender roles regarding the expression of emotion are institutionalized in society. [2%]

Correct

82% answered correctly

Explanation:

Major theoretical approaches to sociology			
Theory	Description	Perspective	Theorists
Functionalism	- Society = an organism - Each part of society works to maintain dynamic equilibrium (homeostasis)	Macro	Émile Durkheim, Talcott Parsons
Conflict	- Society = struggle for limited resources - Inequality based on social class	Macro	Karl Marx, Max Weber
Social constructionism	- Social actors define what is real - Knowledge about world based on interactions	Macro or micro	(Not critical for exam)
Symbolic interactionism	- Meaning & value attached to symbols - Individual interactions based on these symbols	Micro	Charles Cooley, George Herbert Mead
Rational choice/social exchange	Individual behaviors & interactions attempt to maximize personal gain & minimize personal cost	Micro	(Not critical for exam)
Feminist	Examines gender inequality in society	Macro or micro	(Not critical for exam)

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In sociology, **feminist theory** attempts to explain **differences in power** between men and women in society, known as **gender inequality**. Feminist theory includes both macro- and micro-level approaches.

At the macro level, feminist theory considers how *social structures* maintain gender inequality (eg, institutionalized gender discrimination results in the under-representation of women in surgery and engineering). At the micro level, feminist theory considers how *one-on-one interactions* also maintain gender inequality by objectifying or devaluing women (eg, when patients see a female surgeon in scrubs they assume she's a nurse rather than a doctor).

Structural functionalism is a macro-level perspective positing that social structures (such as emotional norms) serve the function in society of maintaining balance or equilibrium. Structural functionalism (not feminist theory) would suggest that differences in the emotional expression of men and women maintain equilibrium in society.

(Choice A) By focusing on the differential treatment of boys (compared to girls) in society (ie, boys are encouraged to suppress their emotions), the statement that strict gender roles are oppressive to boys is an interpretation that *is* consistent with the feminist perspective.

(Choice B) The interpretation that women are often penalized for expressing certain emotions (such as anger) and men are rewarded for those same emotions focuses on how gender inequality in society negatively impacts women, which *is* consistent with the feminist perspective.

(Choice D) The differential socialization of boys and girls in terms of emotional expression (described in Paragraph 4) reinforces traditional gender roles that are institutionalized (ie, established) in society, an interpretation that *is* consistent with the feminist perspective.

Educational objective:

Feminist theory is a macro- and micro-level sociological perspective that attempts to explain the differences in power between men and women (gender inequality) that occur as a result of discriminatory and oppressive small-scale (one-on-one) interactions and large-scale social processes.

Time spent: 0 sec

Subject:

Behavioral Sciences

QID: 401026

System:

Motivation, Emotion, Attitudes, Personality, and Stress

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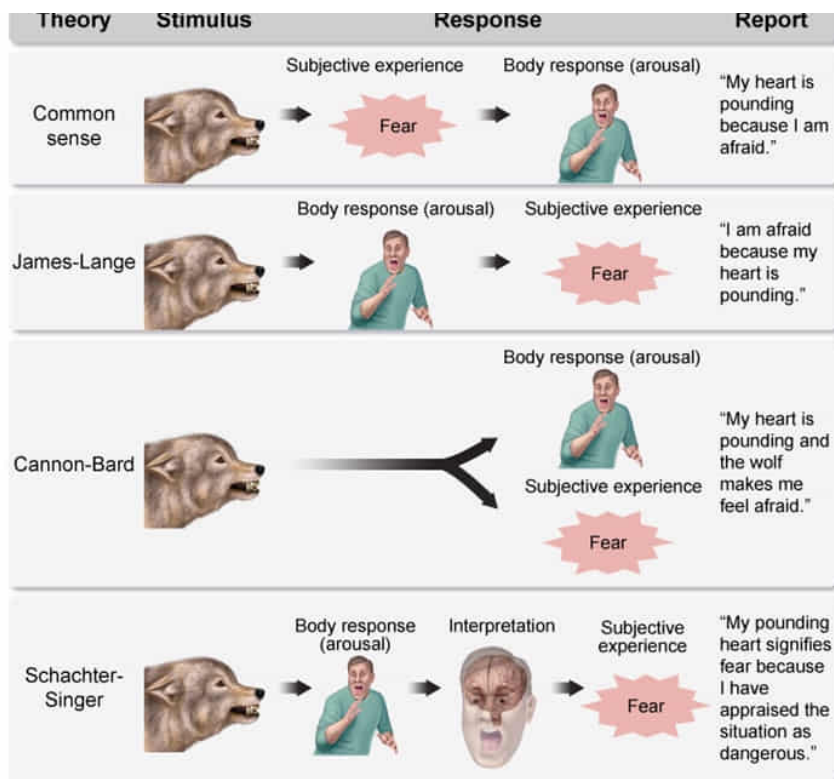
Unlike the descriptions of emotional processing in the second paragraph, the Cannon-Bard theory suggests that:

- ☐ A. emotions do not require cognitive appraisal to be experienced. [3%]
- ☐ B. emotional expressions are innate, not learned. [2%]
- ☐ C. cognition must first occur for emotion to be felt. [9%]
- ✓ ☒ D. physiological arousal and emotion occur simultaneously. [84%]

Correct

85% answered correctly

Explanation:



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The **Cannon-Bard theory** of emotion suggests that physiological **arousal** and **emotion** are separate and independent and therefore **occur simultaneously**. For example, on seeing a wolf, an individual experiences physiological arousal (eg, increased heart rate) *at the same time* that the brain registers the emotion of fear.

Cannon and Bard conducted experiments on cats in which they severed the sympathetic afferent neurons to the brain and then exposed the cats to fear-inducing stimuli. If emotion is caused by physiological arousal (James-Lange theory), then the cats should not demonstrate fear without autonomic feedback from their bodies. Since they did demonstrate fear, Cannon and Bard theorized that the two responses occurred simultaneously and independently.

The Cannon-Bard theory of emotion would explain the events described in the second paragraph by arguing that physiological arousal and the experience of emotion all occur

simultaneously.

(Choice A) Neither the James-Lange nor the Cannon-Bard theory of emotion takes cognitive appraisal into account. The suggestion that emotion requires cognitive appraisal to be experienced is consistent with the Schachter-Singer theory of emotion, which suggests, like the James-Lange theory, that physiological arousal precedes emotion but also suggests that *cognitive interpretation of the situation* allows the correct labeling of the emotion by the brain.

(Choice B) Neither the James-Lange nor the Cannon-Bard theory deals with the innateness of emotion. The idea that emotional expressions are innate and unlearned is consistent with the evolutionary theory of emotion.

(Choice C) The idea that cognition must first occur for emotion to be felt is consistent with the cognitive appraisal theory of emotion, not the Cannon-Bard theory.

Educational objective:

According to the Cannon-Bard theory of emotion, physiological arousal and the experience of emotion occur simultaneously and independently.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401027

System:
Motivation, Emotion, Attitudes,
Personality, and Stress